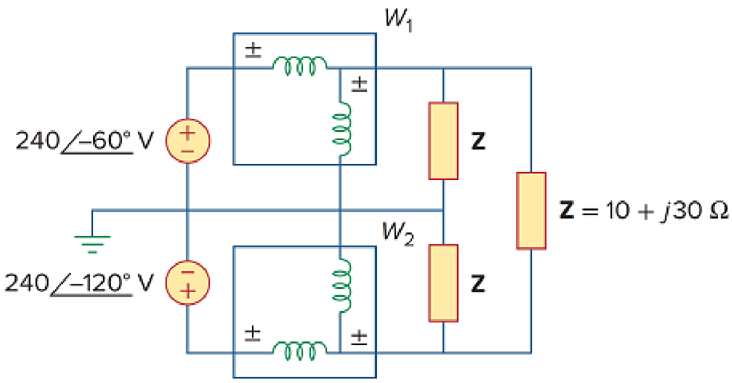
Problem

For the circuit displayed in Fig. 12.74, find the wattmeter readings.

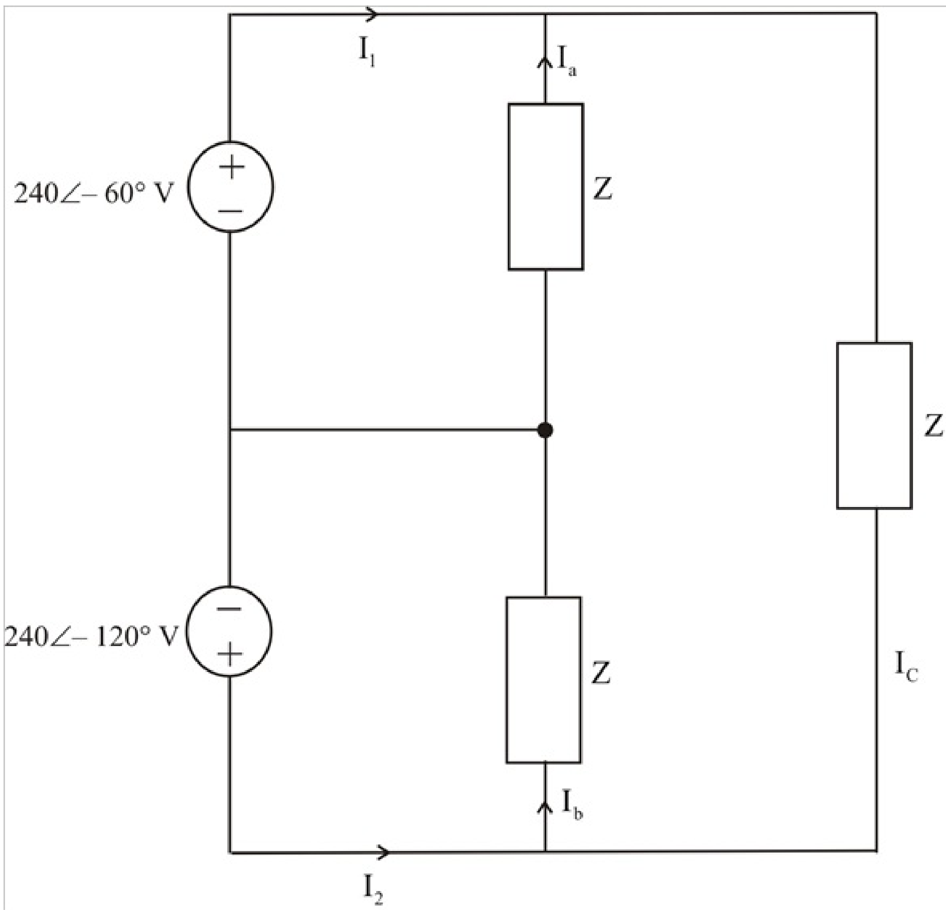
Figure 12.74



Step-by-step solution

Step 1 of 3

Consider the circuit shown below.



Step 2 of 3

$$Z = 10 + j30 = 31.62\angle 71.57°$$

$$I_a = \frac{240\angle -60°}{31.62\angle 71.57°} = 7.59\angle -131.57°$$

$$I_b = \frac{240\angle -120°}{3.62\angle 71.57°} = 7.59\angle -191.57°$$

Step 3 of 3

$$I_C Z + 240\angle -60° - 240\angle -120° = 0$$

$$I_C = I_c = \frac{-240}{31.62\angle 71.570} = 7.59\angle 108.43°$$

$$I_1 = I_a - I_c = 13.146\angle -101.57°$$

$$I_2 = I_b - I_c = 13.146\angle 138.43°$$

$$P_1 = R_e\left[V_1 I_1^*\right]$$

$$= R_e\left[(240\angle -60°)\ (13.146\angle 101.57°)\right]$$

$$= 2360\ W$$

$$P_2 = R_e\left[V_2 I_2^*\right]$$

$$= R_e\left[(240\angle -120°)\ (13.146\angle -138.43°)\right]$$

$$= -632.8\ W\ .$$